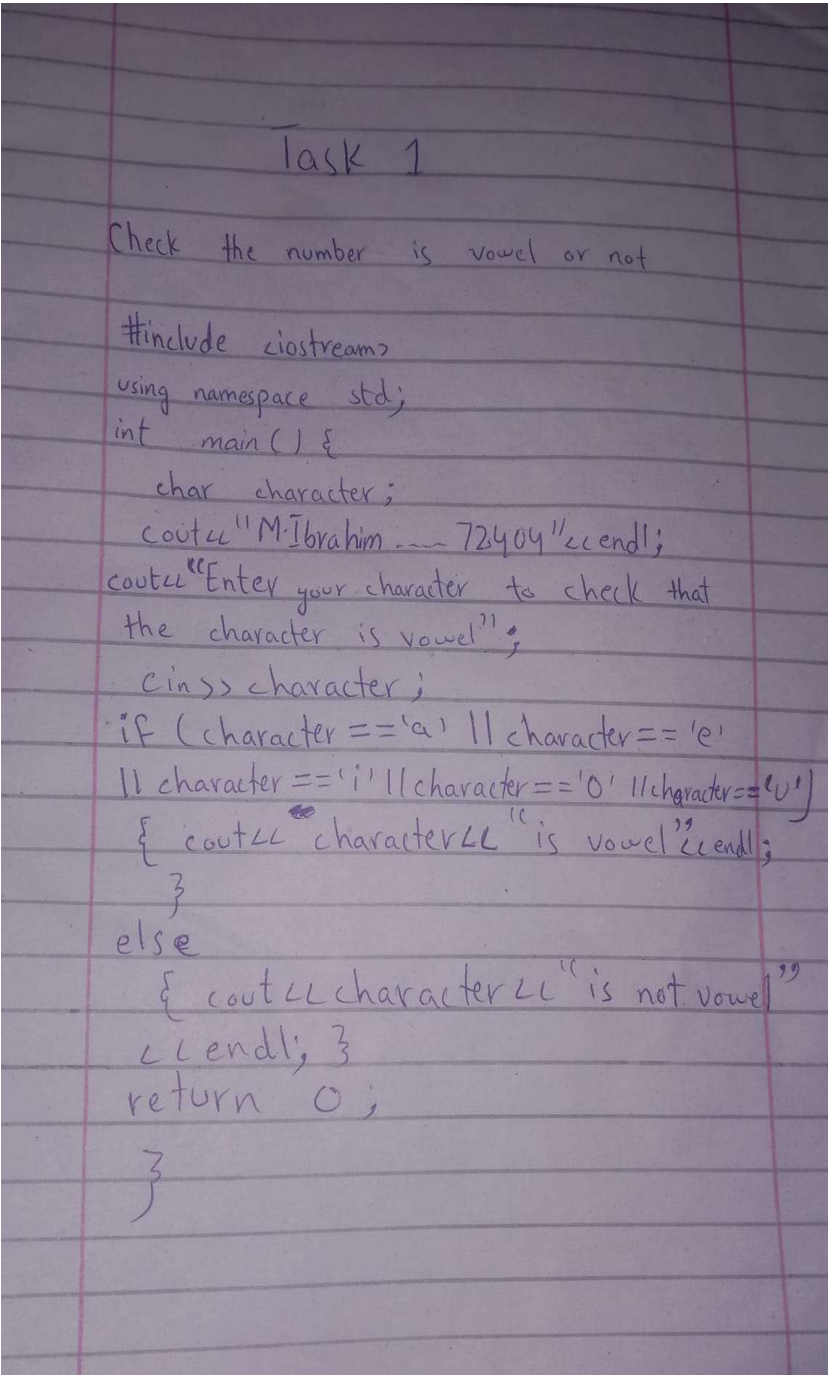
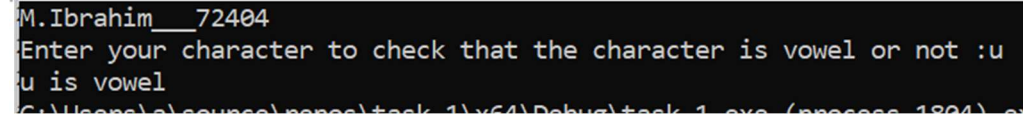


Task #1

```
#include <iostream>
using namespace std;
int main() {
    char character;
    cout << "M.Ibrahim___72404" << endl;
    cout << "Enter your character to check that the character is vowel or not :";
    cin >> character;

    if (character == 'a' || character == 'i' || character == 'e' || character == 'o'
        || character == 'u')
    {
        cout << character << " is vowel";
    }
    else
    {
        cout << character << " is not vowel" << endl;
    };
    return 0;
}
```

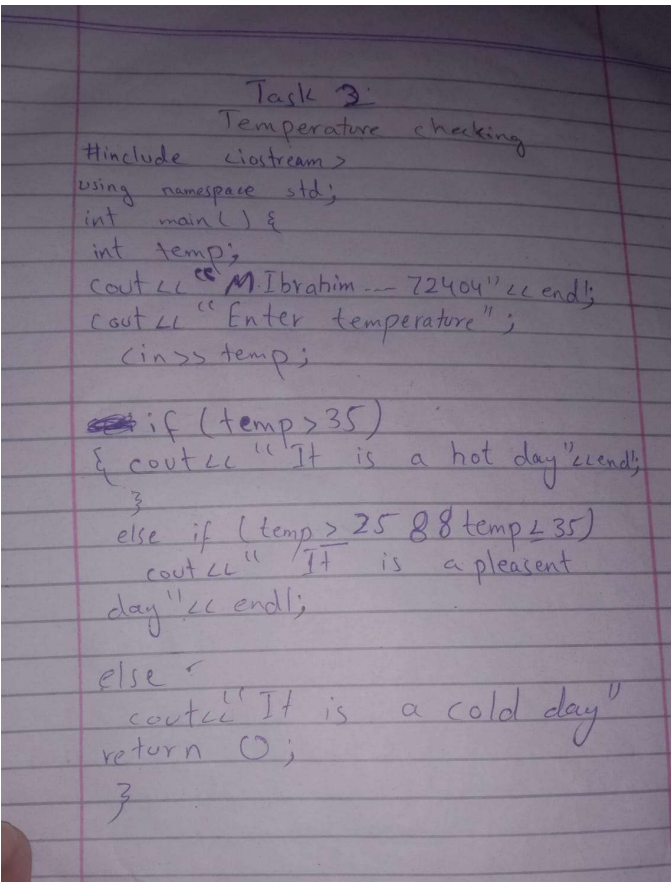
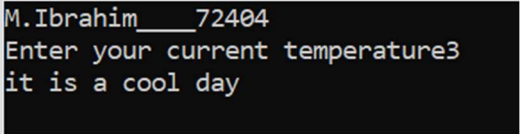
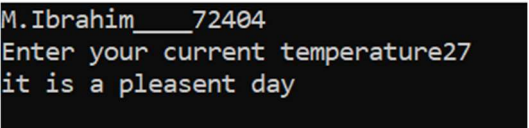
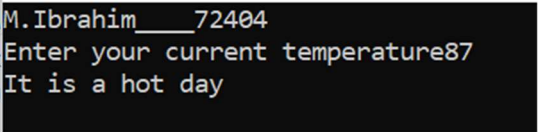


Task#2

```
#include <iostream>
using namespace std;

int main()
{
    int temp;

    cout<<"M.Ibrahim___72404"<<endl;
    cout << "Enter your temperature :";
    cin >> temp;
    if (temp > 35)
    {
        cout << "It is a hot day " << endl;
    }
    else if (temp >= 25 && temp <= 35)
        cout << "It is a pleasent day " << endl;
    else
        cout << "It is a cool day" << endl;
    return 0;
}
```



Task 3

```
#include <iostream>
using namespace std;

int main()
{
    char grade;

    cout << "M.Ibrahim ____72404"<<endl;
    cout << "Enter your grades for testing scores :";
    cin >> grade;
    switch (grade)
    {
        case 'A':case 'a':
            cout << "Your test score is >=90" << endl;
            break;

        case 'B':case 'b':
            cout << "Your test score is 80-89" << endl;
        case 'C':case 'c':
            cout << "Your test score is 70-79 " << endl;
            break; case 'D':case 'd':
            cout << "Your test score is 60-70" << endl;
            break; case 'F':case 'F':
            cout << "Your test score is below 60" << endl;
            break;
        default:
            cout << "invalid grade" << endl;
    }
    return 0;
}
```

```
M.Ibrahim__72404
Enter your grade :a
Your test score is >=90
```

Task 3
Test score

```
#include <iostream>
using namespace std;
int main() {
    char grade;
    cout << "Enter your grade:";
    cin >> grade;
    switch (grade)
    {
        case 'A':case 'a':
            cout << "Your test score is >= 90" << endl;
            break;
        case 'B':case 'b':
            cout << "Your test score is 80-90" << endl;
            break;
        case 'C':case 'c':
            cout << "Your test score is 70-79" << endl; break;
        case 'D':case 'd':
            cout << "Your test score is 60-69" << endl;
            break;
        case 'F':case 'f':
            cout << "Test score is below 60" << endl;
            break;
        default:
            cout << "invalid grade" << endl;
    }
    return 0;
}
```

Task 5

```
#include <iostream>
using namespace std;
int main()
{
    int totalseconds, hours, minutes, seconds;

    cout << "M.Ibrahim ____72404" << endl;
    cout << "Enter number of seconds :";
    cin >> totalseconds;

    hours = totalseconds / 3600;
    minutes = (totalseconds % 3600) / 60;
    seconds = totalseconds % 60;
    cout << hours << " hrs " << minutes << " min " << seconds << "sec" << endl;
    return 0;
}
```

```
M.Ibrahim__72404
Enter number of seconds :6754
1 hrs 52 min 34sec
```

Task 5
display hours mins sec

```
#include <iostream>
using namespace std;
int main() {
    int totalseconds, minutes, hours, seconds;

    cout << "M.Ibrahim__72404" << endl;
    cout << "Enter number of seconds:";
    cin >> totalseconds;

    hour = totalseconds / 3600;
    minutes = (totalseconds % 3600) / 60;
    seconds = totalseconds % 60;

    cout << hours << " hrs " << minutes << " min " << seconds << "sec" << endl;
    return 0;
}
```


Task 4

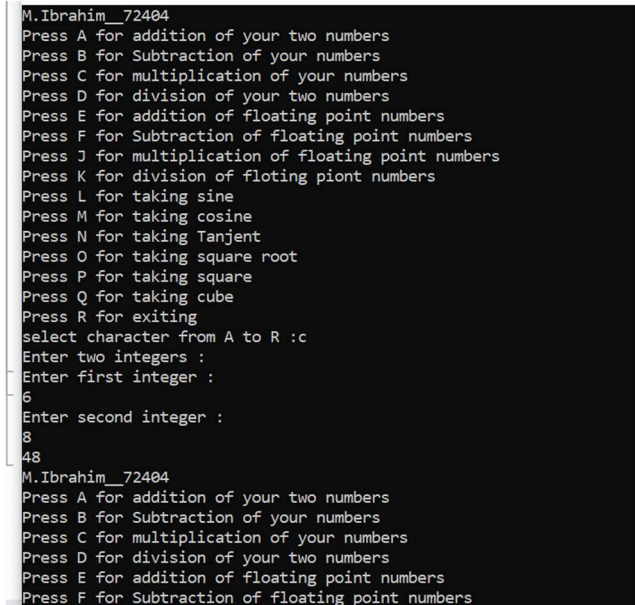
Calculator

```
#include <iostream>
#include <math.h>
using namespace std;
int main()
{
    int num1, num2, value;
    char choise;float f1, f2;double angle;

    do {
        cout << "M.Ibrahim_72404" << endl;
        cout << "Press A for addition of your two numbers" << endl;
        cout << "Press B for Subtraction of your numbers" << endl;
        cout << "Press C for multiplication of your numbers" << endl;
        cout << "Press D for division of your two numbers" << endl;
        cout << "Press E for addition of floating point numbers" << endl;
        cout << "Press F for Subtraction of floating point numbers" << endl;
        cout << "Press J for multiplication of floating point numbers" << endl;
        cout << "Press K for division of floting piont numbers" << endl;
        cout << "Press L for taking sine" << endl;
        cout << "Press M for taking cosine" << endl;
        cout << "Press N for taking Tanjent" << endl;
        cout << "Press O for taking square root " << endl;
        cout << "Press P for taking square" << endl;
        cout << "Press Q for taking cube" << endl;
        cout << "Press R for exiting" << endl;
        cout << "select character from A to R :";
        cin >> choise;

    }

    case'A':case 'a':
        cout << "Enter two integers" << endl;
        cout << "Enter first integer :" << endl;
        cin >> num1;
        cout << "Enter second integer :" << endl;
        cin >> num2;
        cout << "Ans =" << num1 + num2 << endl;
        break;
    case'B':case 'b':
        cout << "Enter two integers" << endl;
        cout << "Enter first integer :" << endl;
        cin >> num1;
        cout << "Enter second integer :" << endl;
        cin >> num2;
    break; case'C':case 'c':
        cout << "Enter two integers :" << endl;
        cout << "Enter first integer :" << endl;
        cin >> num1;
        cout << "Enter second integer :" << endl;
        cin >> num2;
        cout << num1 * num2 << endl;
    break; case'D':case 'd':
        cout << "Enter two integers :" << endl;
        cout << "Enter first integer :" << endl;
        cin >> num1;
```



```
cout << "Enter second integer :" << endl;
cin >> num2;
if (num2 != 0)
    cout << "Ans =" << num1 / num2 << endl;
else
    cout << "error denominatoer is zero :" << endl;
break; case'E':case 'e':
    cout << "Enter two Floating point numbers :" << endl;
    cout << "Enter first float number :" << endl;
    cin >> f1;
    cout << "Enter second float number :" << endl;
    cin >> f2;

    cout << "Ans =" << f1 + f2 << endl;
break; case'F':case 'f':
    cout << "Enter two Floating point numbers " << endl;
    cout << "Enter first float number :" << endl;
    cin >> f1;
    cout << "Enter second float number :" << endl;
    cin >> f2;
    cout << "Ans =" << f1 - f2 << endl;
break; case'j':case 'J':
    cout << "Enter two Floating point numbers :" << endl;
    cout << "Enter first float number :" << endl;
    cin >> f1;

    cout << "Enter second float number :" << endl;
    cin >> f2;
    cout << "Ans =" << f1 - f2 << endl;
break; case'j':case 'J':
    cout << "Enter two Floating point numbers :" << endl;
    cout << "Enter first float number :" << endl;
    cin >> f1;
    cout << "Enter second float number :" << endl;
    cin >> f2;
    cout << "Ans =" << f1 * f2 << endl;
break; case'K':case 'k':
    cout << "Enter two Floating point numbers " << endl;
    cout << "Enter first float number :" << endl;
    cin >> f1;
    cout << "Enter second float number :" << endl;
    cin >> f2;

    if (f2 != 0)
        cout << "Ans =" << f1 / f2 << endl;
    else
        cout << "error denominatoer is zero " << endl;
break; case'L':case 'l':
    cout << "Enter angle in radian :" << endl;
    cin >> angle;
    cout << "Ans=" << sin(angle) << endl;
    break;

    break;
case'M':case 'm':
    cout << "Enter angle in radian :" << endl;
    cin >> angle;
    cout << "Ans =" << cos(angle) << endl;
break; case'n':case 'N':
    cout << "Enter angle in radian :" << endl;
    cin >> angle;
    cout << "Ans =" << tan(angle) << endl;
break; case'o':case 'O':
    cout << "Enter your value : ";
    cin >> value;
    if (value >= 0)
        cout << "Ans =" << sqrt(value) << endl;

    else
        cout << "error: invalid number :" << endl;

break; case'p':case 'P':
    cout << "Enter your value : ";
    cin >> value;
    cout << "Ans =" << pow(value, 2) << endl;
break; case'Q':case 'q':
    cout << "Enter your value : ";
    cin >> value;
    cout << "Ans= " << pow(value, 3) << endl;

    cout << "Ans= " << pow(value, 3) << endl;
    break;

    case 'R':case'r':
        cout << "exiting program...";
        ;
        break;
default:
    cout << "Invalid character" << endl;
}
} while (choise != 'r' && choise != 'R');
return 0;

}
```